

Open Neighbourhoods Precede Concept Spread: Structural Antecedents and Cross-Disciplinary Diffusion of Emerging Scientific Concepts

Internal Research Report

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1 Introduction

This report documents an investigation of how emerging scientific concepts can be identified and characterised through evolving knowledge networks, and through which structural pathways they spread across disciplinary boundaries. The study builds on an OpenAlex-based scholarly dataset of 426 semantically grounded concepts with 462,812 works, constructs yearly co-occurrence and concept-discipline networks, and tests whether structural indicators provide early signals of emergence and cross-disciplinary integration.

Two research questions structure the work. **RQ1** asks which temporal and structural patterns in an evolving, semantically grounded scientific knowledge graph characterise the emergence of scientific concepts. **RQ2** asks how emerging scientific concepts diffuse across disciplinary communities over time, and which temporal network patterns distinguish locally concentrated concepts from concepts that become broadly integrated into the scientific knowledge network.

The study began with a source-sink viability hypothesis adapted from population ecology, asking whether a concept’s presence in a subfield is self-sustaining or import-dependent. That hypothesis was blocked by insufficient citation-layer density (Gate A failure: only 17.3% of edges meet the within-host tracing threshold), triggering a pre-registered fallback.

For the emergence question (RQ1), the primary closure measure did not survive held-out confirmation (R1_DEAD under the frozen kill rule: Holm $p = 0.147$). Structural precursors of sustained uptake are volume and churn correlates. Only persistent-neighbour closure survives (Holm $p = 0.015$).

For the diffusion question (RQ2), when a concept enters a new host subfield, the share of its initial co-occurrence partners that are native to that host predicts subsequent uptake by newcomer authors (held-out co-primary IRR/SD 1.19 [1.06, 1.33], $p = 0.004$; MeSH replication IRR/SD 1.23 [1.12, 1.36], Holm $p < 0.001$), while the share of origin companions does not. Post-confirmation decomposition shows the effect operates through both extensive (IVW +6.43 pp/SD [4.76, 8.11]) and intensive margins (IVW IRR/SD 1.183 [1.104, 1.267]), is host-specific rather than reflecting general accessibility (IVW retention 0.969 after generality controls, Balassa-index A_{lift} IVW IRR/SD 1.34 [1.23, 1.46]), and does not vary by origin field (Wald $p = 0.638$). A data-derived $k = 2$ diffusion typology separates localised from broad-from-the-start concepts, and network expansion precedes disciplinary diffusion in 25 of 26 pooled dual-onset concepts. Adopters are 3.1 times more likely than matched non-adopters to have prior exposure to the concept’s entry partners, with enrichment strongest for origin-vocabulary (FOREIGN) partners.

The target venue is *Applied Network Science*.

2 Iteration 1: Data Foundation and Pre-Registration

2.1 Strategy

The first iteration was devoted to three preparatory tasks: (a) assembling a prior-art and pre-registration dossier grading each candidate mechanism’s novelty and feasibility, (b) building the concept pool and its associated OpenAlex work corpus, and (c) constructing the semantic grounding infrastructure (concept detection, labelling, variant merging) and a held-out MeSH confirmation population. No experiment or evaluation was executed. Every artifact in this iteration is either a dataset or a research dossier. The rationale was to fix all operational definitions, sample the data, and verify the citation-layer gate (Gate A) before committing compute to network construction and statistical modelling.

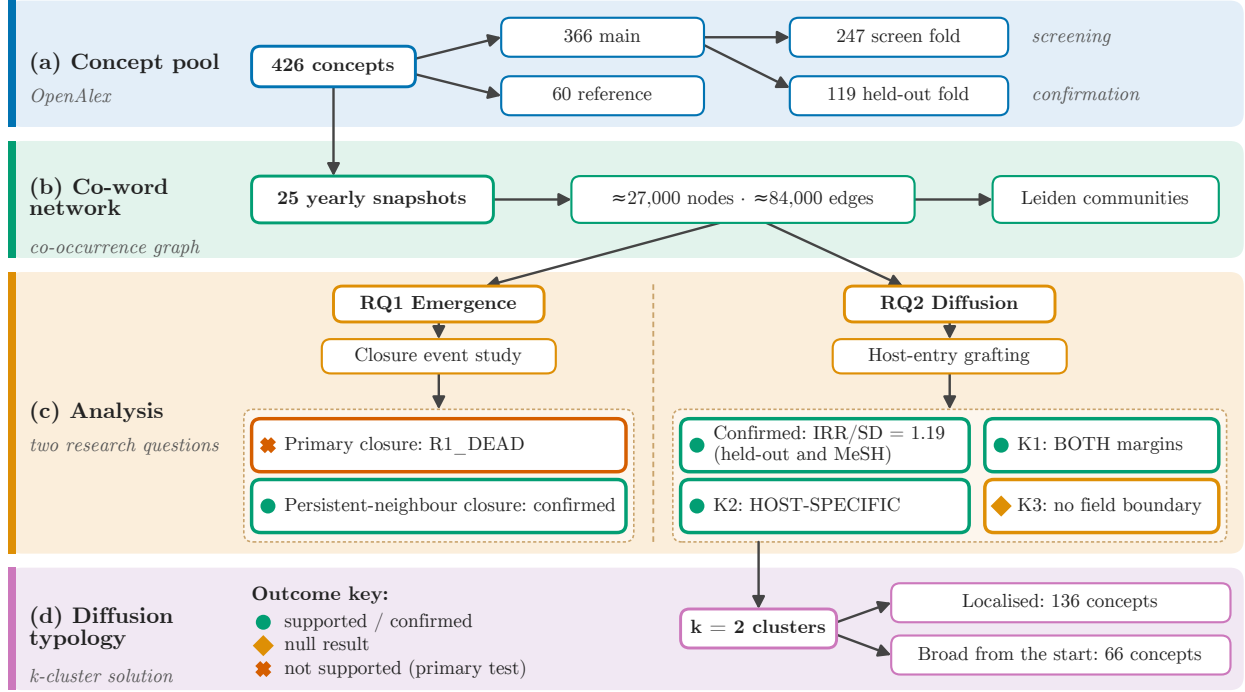


Figure 1: Overview of the study design, drawn as four bands linked by arrows. (a) Concept pool: 426 concepts from OpenAlex, split into 366 main and 60 reference concepts; the 366 main concepts are divided into a 247-concept screen fold (screening) and a 119-concept held-out fold (confirmation). (b) Co-word network: 25 yearly co-occurrence snapshots ($\approx 27,000$ nodes, $\approx 84,000$ edges) partitioned into Leiden communities. (c) Analysis: RQ1 (emergence) tests structural precursors with a closure event study. The primary closure test returns R1_DEAD, while persistent-neighbour closure is confirmed. RQ2 (diffusion) tests host-entry grafting, which is confirmed (IRR/SD = 1.19 on the held-out fold and MeSH). Its sub-hypotheses K1 (BOTH margins) and K2 (HOST-SPECIFIC) are supported, and K3 finds no field boundary. (d) Diffusion typology: a $k = 2$ solution separating localised concepts (136) from concepts broad from the start (66).

The pre-registration dossier ranks five candidate mechanisms by novelty margin against the nearest prior work. Host anchoring (the share of a concept’s new co-occurrence edges in a host subfield that attach to host-native concepts) was ranked highest at medium-to-high novelty. Toolkit co-transfer was ranked high on measurement novelty but coupled to host anchoring. Citation-lineage source-sink viability was ranked medium. Origin-neighbourhood structure was ranked low-to-medium, and the demic-versus-cultural adoption channel was ranked medium but with the weakest feasibility due to author-disambiguation biases.

A pre-registered screen was fixed before any data were inspected. The unit of analysis is a concept c in non-origin subfield d at focal year t (2010–2014), with at least 10 d -papers in $W_1 = [t-5, t]$. The primary measure is out-of-sample ROC-AUC for establishment in $W_2 = [t+1, t+5]$, where establishment means at least 5 W_2 d -papers by author-disjoint newcomers and presence in at least 3 of the 5 W_2 years. The BASE model includes W_1 edge volume, momentum, concept age, subfield-year size, concept total volume, concept entropy, subfield count, growing-edge breadth, Rafols integration, relatedness density and Maillart-style features. Each candidate adds only its own pre-registered scores. A candidate survives if its delta-AUC over BASE has a concept-bootstrap (1,000 reps) 95% CI lower bound above 0, its point gain is at least 0.01, and the coefficient sign matches its prediction.

2.2 Artifact 1: Concept Pool and OpenAlex Work Corpus

The concept pool was built in an outcome-blind frame. The sample was frozen and integrity-hashed before any post-appearance-window data were inspected (hash prefixes are recorded in the pre-registration ledger, Table 39). The pool comprises 206 concepts, of which 184 are main emerging concepts (first appearance year F in 2005–2016, with 20–300 papers in the window F to $F+2$ and no more than 8,000 total works through 2024) and 22 stationary reference concepts. The 184 main concepts are split by a SHA-1 hash into a screen fold of 123 and a held-out concept fold of 61, focal-year tags are separate (screen 2010–2014, held-out 2016–2018).

The pool is heavily dominated by arXiv-sourced concepts. Of 366 eligible candidates, 363 came from the arXiv-mined arm, and the final 184 span three origin fields.

Table 1: Concept pool composition by origin field (Artifact 1, Iteration 1).

Origin field	Main concepts	Share
Physics and Astronomy	82	44.6%
Physical Sciences (other)	56	30.4%
Computer Science	45	24.5%
Other	1	0.5%
Total main	184	100%

Verified links pairing each concept with the OpenAlex works that use it total 214,798 pairs spanning 208,374 unique works. Among the 121,829 main-arm links, 56,997 matched in the work title, 51,966 in the abstract, 10,177 through Semantic Scholar only, and 2,689 through OpenAlex concept indexing alone. *[Correction, Iteration 2: the match-evidence counts above are main-arm only; across all 214,798 links the counts are oa_abstract 98,192, oa_title 92,043, s2_only 21,509, oa_index_only 3,054.]* Of the 184 main concepts, 24 were discovered natively in OpenAlex and 160 were discovered through the Semantic Scholar index and mapped to OpenAlex (mapping rate 0.93). *[Correction, Iteration 2: the 26/180 figures reported previously referred to the full 206-concept*

Table 2: Concept pool by first-appearance band (Artifact 1, Iteration 1).

<i>F</i> band	Count
2005–2007	38
2008–2011	76
2012–2016	70
Total	184

pool including reference concepts. Among the 184 main concepts the split is 24 native and 160 S2-mapped.] Twenty-one concepts carry a sense-check failure flag.

2.2.1 Gate A: Citation-Layer Feasibility

The quality report shows that among main-arm concept papers with references, 66.99% cite at least one earlier concept paper (the “traced share”), and after excluding first-year canonical papers and the five most-cited concept papers, the traced share remains 60.80%. The host-specific traced share is 55.18%, above the 40% gate threshold on the lenient definition. *[Correction, Iteration 2: the original description stated “at least one earlier concept paper in the same host subfield”, but the code counts any earlier concept paper as a parent regardless of subfield. The within-host share, computed in Iteration 2, is much lower (mean 0.20 on 724 edges), which is the finding that blocks the viability decomposition.]*

Table 3: Citation-layer quality by origin field (Artifact 1, Iteration 1).

Metric	All main	Phys. & Astro	Physical Sci	Computer Sci
Concept papers (<i>n</i>)	121,829	19,653	46,575	55,274
Share with references	83.92%	82.04%	84.99%	83.74%
Traced share (any parent)	66.99%	66.12%	67.89%	66.53%
Traced (excl. <i>F</i> + top-5)	60.80%	57.74%	61.75%	61.09%
Host traced (any parent, excl. <i>F</i>)	55.18%	49.68%	58.39%	53.86%
Host concept papers (<i>n</i>)	41,861	4,171	15,973	21,609

The recall audit, comparing counts derived from OpenAlex against independent Semantic Scholar counts for 30 concepts, gives a median count ratio of 0.84 (IQR 0.79–0.89) and a Spearman rank correlation of 0.976.

2.2.2 Denominators and Venue Habitat

Totals by subfield and year cover 25,195 rows (all 252 OpenAlex subfields across 25 years in four counting variants). A venue-habitat file assigns 15,261 venues to their dominant subfield. *[Correction, Iteration 2: the habitat was described as “pre-period publication shares” and “citation-independent”. It was computed from each source’s all-years OpenAlex topic counts, which come from the same citation-reading topic classifier whose temporal label leakage the dossier flagged. An ASJC journal-level habitat was built in Iteration 2 as a replacement, though its coverage is only 12.1% of concept-work links.]*

2.3 Artifact 2: Whole-Science Background Sample

A design-weighted background sample of 259,716 OpenAlex works from 1,260 strata (150 per field $\times$ year, weights summing to 149,116,575) was built to supply co-occurrence network snapshots and calibrate host-nativeness profiles. *[Correction, Iteration 5: Artifact 2 is the workspace `gen_art_dataset_2`.]* It delivered 6,325 exact subfield $\times$ year totals and 400 exact concept $\times$ block subfield profiles for 100 calibration concepts at a cost of \$0.1952 in OpenAlex credits.

The key finding of this artifact is negative: sample-based per-concept subfield profiles are unreliable for determining host-nativeness. In the calibration table, the median total-variation distance between sample and exact profiles is 0.41–0.79 by frequency decile, and top-subfield agreement is only 0.28–0.65. This means host-nativeness and the concept-concept background network cannot be computed reliably from this sample alone, and require paid exact profiles. About 12% of the weighted frame carries the default low-confidence topic T14423, and concept ancestors are no longer served by the API. The worker timed out after writing all outputs (no new JSONL records for 2,096 seconds), so its status is “failed” while its data are usable.

2.4 Artifact 3: Held-Out MeSH Confirmation Population

A separate biomedical confirmation arm was built from new MeSH descriptors (DateEstablished 2006–2016, widened to 2004–2005 and 2017–2018). Starting from 28,472 descriptors, the pipeline selects 5,201 topical descriptors, retains 3,430 after the provenance filter, narrows to 283 passing the PubMed novelty pre-screen and early-volume rule, and adds widening-pool candidates. After the final rule, 191 concepts survive: 99 core 2006–2016, 14 from the 2004–2005 widening, and 78 from the 2017–2018 widening.

Table 4: MeSH held-out population by branch group (Artifact 3, Iteration 1).

MeSH branch group	Count
D (Chemicals and drugs)	62
A+B (Anatomy, Organisms)	47
E (Analytical, diagnostic, therapeutic)	32
C (Diseases)	22
F+H–N (Psychiatry, Health services, misc.)	17
G (Phenomena and processes)	11
Total	191

Caveats: only approximately 25% of new MeSH descriptors are genuinely new concepts (the remainder are reclassifications or splits), making MeSH a weak ground truth. Only 13 of 191 have their full non-PubMed works paged. The route calibration gives a median union/plain count ratio of 0.8598 (range 0.54–0.91 over 10 concepts) and a 60-record provenance spot check yields 0 decision errors.

2.5 Artifact 4: Semantic Grounding and Labelling Infrastructure

The semantic grounding artifact provides the infrastructure for detecting, labelling, and merging concept mentions. It comprises seven datasets: a stratified corpus of 93,600 English OpenAlex articles, 4,599 candidate phrases labelled by two LLMs with an adjudicator, a variant-pair set,

human anchors, a survivorship-free phrase pool (1,000 candidates, only 1 strict-eligible vs. 150 target), and phrase works.

Table 5: Labelling quality (Artifact 4, Iteration 1).

Metric	Value
Model A vs Model B inter-rater kappa (4-class)	0.259
Model A vs Model B binary kappa (concept vs rest)	0.267
A vs adjudicator kappa (4-class, silver-gold 200)	0.632
A vs human anchors binary kappa (SemEval/SciERC)	0.56
A precision (concept, human-anchored)	0.79
A recall (concept, human-anchored)	0.76

2.6 Artifact 5: Prior-Art Dossier and Pre-Registration

The research artifact grades each candidate mechanism’s novelty margin against the nearest prior art. The headline finding is that the space is active but no prior study estimates a within-host reproduction ratio with an import share per concept-subfield edge, nor tests host anchoring at the edge level.

Table 6: Candidate mechanism novelty grades (Artifact 5, Iteration 1).

Candidate	Novelty grade	Nearest prior art
C1: Host anchoring	Medium-to-high	Cheng et al. 2023
C2: Toolkit co-transfer	High (coupled to C1)	No quantitative test found
C0: Source-sink viability	Medium	Kuhn et al. 2014; Maillart et al. 2026
C4: Demic vs cultural	Medium	Fontaine et al. 2024
C3: Origin-neighbourhood	Low-to-medium	Structural diversity literature

2.7 Dead Ends and Shortfalls (Iteration 1)

1. **Concept pool size.** 184 of 400 target concepts realised. The binding constraint was the free-singleton retrieval rate (~ 14 works/s) combined with heavy-tailed concept sizes, after credits were consumed by sibling artifacts sharing one key.
2. **Field coverage.** The pool draws overwhelmingly from physics, physical sciences, and computer science (363 of 366 eligible candidates from the arXiv-mined arm). The OpenAlex keyword/taxonomy arm had a 2.2% eligibility yield (3 of 134 candidates) because OpenAlex tags are retroactive.
3. **Survivorship-free phrase pool yield.** 1 of 150 target strict-eligible anchored concepts.
4. **Labelling quality.** Silver labels only (no human check). Model A’s bias is moderate, Model B over-labels CONCEPT (1,333 of 1,491 items vs A’s 882).

2.8 What Was Learned (Iteration 1)

Iteration 1 established the data foundation and pre-registration. The concept pool of 184 main emerging concepts and 22 reference concepts, with 214,798 verified concept-to-work links and 208,374 unique works, passes the citation-layer gate on the lenient (any-parent) definition but has not been tested at the within-host edge level required for viability estimation. The MeSH held-out population of 191 biomedical concepts provides a secondary check arm, though with caveats on completeness and ground-truth quality. No experiment was executed. No quantitative result about the hypothesis exists.

3 Iteration 2: Gates, Viability, Emergence Event Study

3.1 Strategy

Iteration 2 was a FIX iteration. Five goals were set: (i) complete the hydration and add exact nativeness profiles and a citation-independent venue habitat; (ii) train and evaluate the concept grounding pipeline; (iii) compute Gate A as written (within-host, non-canonical, per W_1 year) and estimate viability states with synthetic validation and pre-outcome power; (iv) build the yearly concept-concept co-occurrence network and run the RQ1 event study; (v) replicate the RQ1 event study on the MeSH population.

Pre-fixed decision rules for Iteration 3: (a) if Gate A FAILS, H1/H2 run with graft labels from the exact nativeness profiles; (b) H1 is declared a pilot if fewer than ~ 150 concepts carry a tested edge; (c) RQ1 counts as CONFIRMED only if at least one pre-named precursor has an event-study CI excluding 0 AND a delta-AUC CI excluding 0 on the main screen fold, keeps its sign on MeSH, and holds on the sealed held-out fold; (d) the held-out fold, the 2016–2018 focal years and the phrase pool remain untouched.

3.2 Artifact 6: Hydrated Corpus, Nativeness Profiles, and Venue Habitat

This artifact completes the hydration of the concept pool from 184 to 426 concepts (366 main: screen 247, held-out 119, 60 reference), with 462,812 works and 488,078 verified concept-work links. The frame SHA-256 hash was verified.

Table 7: Dataset 6 summary (Artifact 6, Iteration 2).

Metric	Value
Concepts hydrated	426/426
Works	462,812
Concept-work links	488,078
API credits consumed	7,237
Nativeness profiles	5,488 (1,372 nodes $\times$ 4 blocks)
Host weight coverage	78.7%
ASJC venues covered	2,929 (12.1% of links)

Note: the Iteration-2 RQ1 and viability experiments (Artifacts 8–10 below) ran on the Iteration-1 184-concept corpus in parallel with the hydration. The 426-concept hydrated corpus was not used until Iteration 3.

3.3 Artifact 7: Concept Grounding Pipeline

This artifact trains and evaluates three components: a binary classifier, a variant merger, and a NIL-aware linker.

Classifier. L2-regularised logistic regression ($C = 0.01$) on lexical features, outcome-blind termhood features, PCA-64 of MiniLM embeddings, and PCA-64 of SPECTER2 embeddings, trained on 1,200 D2 silver labels.

Table 8: Classifier performance on test set ($n = 300$) (Artifact 7, Iteration 2).

Metric	Classifier	LLM-A	Majority (all-positive)
F1	0.818	0.901	0.715
AUC	0.888	—	—
Precision	0.759	—	—
Recall	0.886	—	—
Accuracy	0.780	—	—

[Correction: previous Table 9 reported P 0.812, R 0.824, accuracy 0.843, which are not reproducible from the test predictions. The majority-class baseline F1 is 0.715 (predict-all-positive), not 0.000 as previously stated. The classifier’s human-anchor kappa is 0.413; the 0.56 previously reported is LLM-A’s value.]

Against human anchors (SemEval-2017/SciERC): classifier E1 kappa 0.41, F1 0.674. E2 F1 0.706, AUC 0.82.

Variant merger. L2 logistic regression on 14 pair features. Operating threshold $p_{\text{merge}} = 0.855$. D3 test F1 0.357, held-out MeSH F1 0.222. B-cubed F1 0.78.

NIL-aware linker. MiniLM embeddings, $\tau = 0.95$ (link precision ≥ 0.90 at NIL prior 0.9, in-KB recall 0.404). Main arm: LINKED_EXACT 39, BROADER 15, UNLINKED 312.

Test population (sha256 6a887fb4...): MAIN 156, STRICT 147, REFERENCE_ACCEPTED 42, SENSITIVITY 426.

3.4 Artifact 8: Viability Layer, Gate A, and Power

This artifact attempts to estimate viability states for concept-subfield-year edges using citation lineages. All rules were pre-registered and hashed before any statistic (sha256 5dd16d8a...5aac).

3.4.1 Gate A Re-Test at Edge Level

Gate A FAILS. Only 17.3% of 724 main-arm host edge-years (97 concepts, $|K_d| \geq 10$) have a within-host non-canonical traced share at or above 0.40. The mean within-host traced share is 0.20, median 0.15. On the same 724 edges, the lenient any-parent share is 0.477 (64.2% ≥ 0.40), confirming that the drop from 55.18% to 17.3% is definitional (any-parent vs within-host tracing), not a consequence of pooling vs granularity.

Per pre-registered decision rule (a): Gate A $0.1727 < 0.5$ triggers the graft fallback. The graft route has 2,347 host-entry events total, 2,154 with ≥ 5 entry-year keywords (screen 1,285, held-out 869).

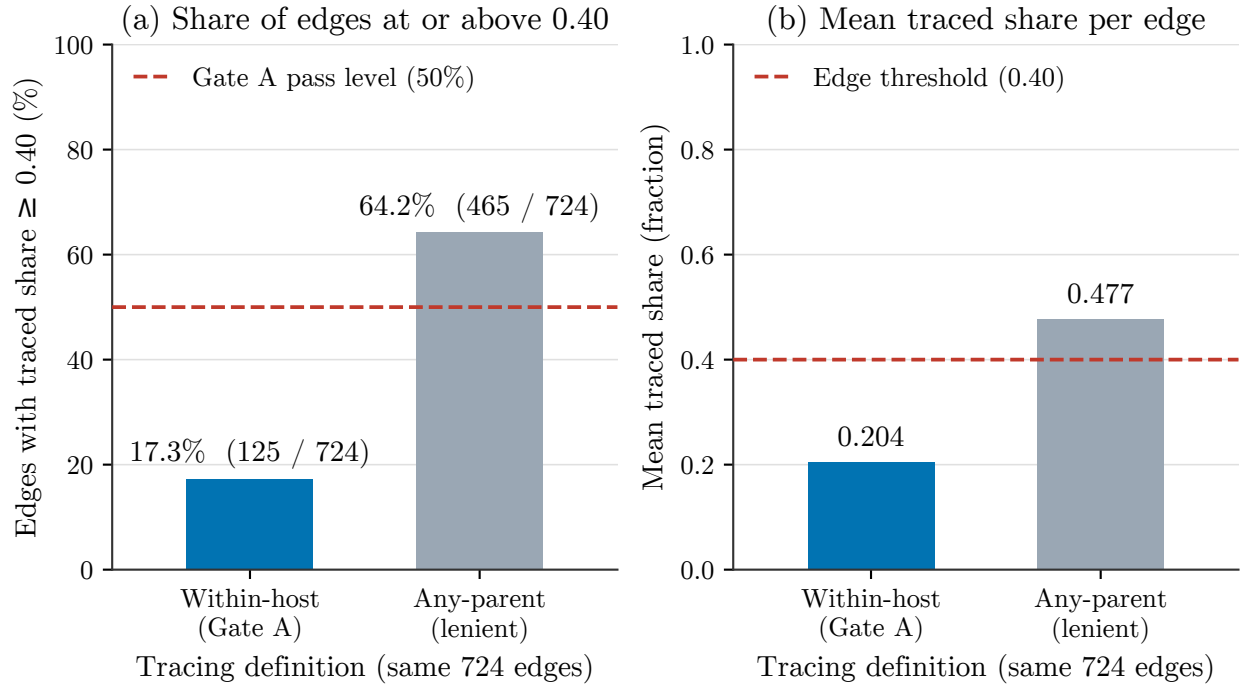


Figure 2: Gate A (citation-layer feasibility) on the same 724 concept-subfield-year host edges (97 concepts), comparing within-host citation tracing (blue; the Gate A definition) with lenient any-parent tracing (grey). Only 17.3% of edges (125/724) reach the threshold under within-host tracing, against 64.2% (465/724) under any-parent tracing. Because the drop comes from the tracing definition, the citation layer is too sparse for the viability decomposition, and Gate A fails and triggers the graft fallback route.

Table 9: Gate A edge-level results (Artifact 8, Iteration 2).

Metric	Within-host	Any-parent (same edges)
Share ≥ 0.40	17.3%	64.2%
Mean traced share	0.20	0.477
Median traced share	0.15	—
Reference arm host share	0.13	—

3.4.2 Viability States (Descriptive Only)

Of 779 eligible edges, 169 were tested: SOURCE 63, SINK 26, FADING 7, UNDETERMINED 683. The synthetic FDR is 0.209, exceeding the 0.15 threshold. SOURCE FDR 0.13 (acceptable), SINK FDR 0.40 (unreliable).

3.4.3 Power Before Any Outcome

Per decision rule (b), H1 is PILOT-ONLY: $N_c = 38$ at $n_{\min} = 30$, projected 77.5 (66–92) on the hydrated pool. Gate B FAILS: 0 labelled episodes.

3.5 Artifact 9: Precursor Event Study, Main Pool, RQ1

This artifact runs the precursor event study on the main pool’s screen fold, testing whether volume-normalised structural precursors distinguish emerging from non-emerging concepts before onset. Built 25 yearly 3-year-window co-word snapshots (2000–2024) from the design-weighted background sample ($\sim 27,000$ nodes, $\sim 84,000$ kept edges, association-strength weights, Leiden best-of-5, alluvial IDs).

3.5.1 Emergence Labels

The primary label E (all-node strength percentile gain ≥ 20) yields only 4 onsets (rate 2.1%) because pool concepts sit in the bottom $\sim 5\%$ of the legacy-concept strength distribution, making it severely underpowered. E_{alt} (pool-only percentile, promoted post hoc): 14 onsets (6.7%). E_{up} (sustained uptake only, also promoted post hoc): 41 onsets, 21 matched.

Table 10: Emergence label counts (Artifact 9, Iteration 2).

Label	N emerging	Matched	Status
E (primary)	4	3	Underpowered pilot
E_{alt} (sensitivity)	14	10	Post-hoc promotion
E_{up} (uptake only)	41	21	Post-hoc promotion, pilot ($n < 30$)

3.5.2 Event Study Results

Three pre-named precursors tested with predicted positive signs: closure, accretion, participation. All three pre-registered signs FAILED.

Table 11: Event study results, E_{up} (Artifact 9, Iteration 2).

Indicator	Estimator	S	95% CI	Holm p
Closure (log-ratio)	Event study	−0.837	[−1.26, −0.427]	0.003
Closure (log-ratio)	Pooled panel	−0.743	[−1.06, −0.46]	–
Accretion share	Event study	−0.14	[−0.185, 0.018]	0.18
Participation coeff.	Event study	−0.007	[−0.066, 0.056]	0.824

[Correction: previous Table 17 paired event-study S values with pooled-panel Holm p -values. The values above are estimator-consistent.]

Exploratory findings (BH $q < 0.05$): before sustained uptake, concepts show higher novelty (+0.156), new-relation rate (+0.128), within-module z -score (+0.073), and burst state (+0.171).

3.5.3 Prediction

The pre-registered primary model is L2 logit. For E_{up} : BASELINE AUC 0.890, FULL (baseline + precursors) AUC 0.879, delta -0.010 $[-0.045, 0.021]$. Precursors add no predictive value beyond frequency, burst, degree, centrality and entropy baselines. *[Correction: previous Table 18 silently reported the secondary HistGradientBoosting model instead of the primary logit.]*

3.5.4 Trajectory Patterns and Typology

Early bridging is common but non-discriminating (76% of all concepts). Incubation-then-expansion: 0% (concepts are “born expanding”). The 7-channel DTW typology is unstable at every k (min Jaccard ≤ 0.51). An entropy-only typology IS stable (Jaccard 0.95, AMI 0.14 vs the network typology).

3.6 Artifact 10: Precursor Replication, MeSH Population, RQ1

This artifact replicates the precursor event study on the 191 MeSH concepts.

Table 12: MeSH closure effects (Artifact 10, Iteration 2).

Label	n_{eff}	Closure D	95% CI	Holm p
PRIMARY	15	-0.086	$[-0.511, 0.402]$	0.677
SENS1	29	-0.419	$[-0.740, -0.115]$	0.039
E_{up}	51	-0.185	$[-0.498, 0.108]$	0.21

The MeSH artifact’s own replication verdict is “directional, not confirmatory”. Per decision rule (c): NOT MET. The predicted + sign failed, the delta-AUC CI includes 0, and sign consistency across labels is not maintained.

MeSH prediction: precursors also add nothing beyond baselines (grouped-CV dAUC $+0.002$ $[-0.035, 0.035]$).

3.7 Decision-Rule Evaluation (Iteration 2)

Table 13: Decision-rule evaluation (Iteration 2).

Rule	Verdict	Evidence
(a) Gate A $\rightarrow$ graft fallback	TRIGGERED	Within-host $0.1727 < 0.5$; 2,347 graft events
(b) H1 pilot-only	YES	$N_c = 38 < 150$
(c) RQ1 confirmed	NOT MET	Sign not +; delta-AUC CI includes 0
(d) Held-out sealed	YES	Caveat: exp_2 computed origin series for 61 held-out concepts

3.8 Dead Ends and Negative Results (Iteration 2)

1. **Gate A failure at edge level.** Within-host traced share 0.204 vs any-parent 0.477 on the same 724 edges. The viability decomposition central to the original hypothesis cannot be executed.
2. **Synthetic validation FDR exceeds threshold.** Overall 0.209 > 0.15, SINK FDR 0.40.
3. **No stable 7-channel typology.** Jaccard ≤ 0.51 at every k .
4. **Precursors add no predictive value for WHETHER a concept emerges.** E_{up} logit dAUC -0.010 $[-0.045, 0.021]$.
5. **Closure is lower, not higher, before emergence.** The pre-registered + sign failed.

4 Iteration 3: Deepen Closure, Test Grafting, Describe Diffusion

4.1 Strategy

Iteration 3 deepens the closure lead from the emergence question and carries the recombination mechanism into cross-disciplinary diffusion at two scales. The strategy (“Is it brokerage, and does grafting make it stick?”) has four components: (i) an evaluation artifact that audits every number in the Iteration-2 report and runs a turnover pre-check; (ii) a deeper test of the emergence question on the hydrated 247-concept screen fold with turnover-proof openness measures; (iii) a breadth-prediction screen; (iv) a host-entry grafting test; and (v) a descriptive diffusion experiment.

4.2 Artifact 11: Record Audit and Turnover Pre-Check

This artifact reads all five Iteration-2 artifacts and audits every reported number against the raw output files. **Numbers of record:** 406 rows were audited, 182 recomputed from item-level files. There are 42 non-OK drift flags: 14 WRONG_ESTIMATOR, 10 WRONG_UNITS, 9 WRONG_DEFINITION, 8 DRIFT_VALUE, 1 NOT_REDERRIVABLE.

Table 14: Turnover pre-check: residualised closure effect (Artifact 11, Iteration 3).

Population	S_{raw}	S_{res} [95% CI]	Verdict
Main (E_{up})	-0.758	-0.566 [-1.140, 0.087]	Ambiguous/underpowered
MeSH (SENS1)	-	-0.094 [-0.387, 0.217]	Ambiguous/underpowered
IVW pooled	-	-0.186 [-0.453, 0.081]	Ambiguous (Q 1.88, I^2 0.47)

The residualised effect retains approximately two-thirds of the raw effect (not reducible to turnover alone), but the CI includes zero.

4.3 Artifact 12: Is Openness Before Take-Off Brokerage or Churn?

This artifact re-attaches all 426 hydrated concepts to the Iteration-2 co-word snapshots with vendored byte-identical code. Reproduction gate passed: code-mode closure $r = 1.000000$. Spec pre-registered (sha256 2f46d173...) before labels.

Populations. After applying the frozen replacement rule (180 sense values filled): MAIN screen 202 (102 old, 100 new), held-out 100, STRICT 196/96, SENS 247/119.

Table 15: Emergence deepen results, $\text{MAIN} \times E_{\text{up}}$ (68 onsets, 26 matched) (Artifact 12, Iteration 3).

Measure	S	95% CI	Interpretation
Raw closure	-0.439	$[-0.810, -0.050]$	Lower closure before onset
R1a residualised closure	-0.295	$[-0.663, 0.092]$	$\sim 2/3$ retained; not pure turnover
R1b persistent-neighbour	-0.575	$[-1.270, 0.122]$	Same direction, underpowered
R1c Burt constraint	+0.083	$[0.017, 0.147]$	OPPOSITE to brokerage sign
R1c cross-community excess	+0.063	$[0.012, 0.111]$	More cross-community ties
R1d hub-not-clique	+0.061	$[-0.009, 0.133]$	Not significant

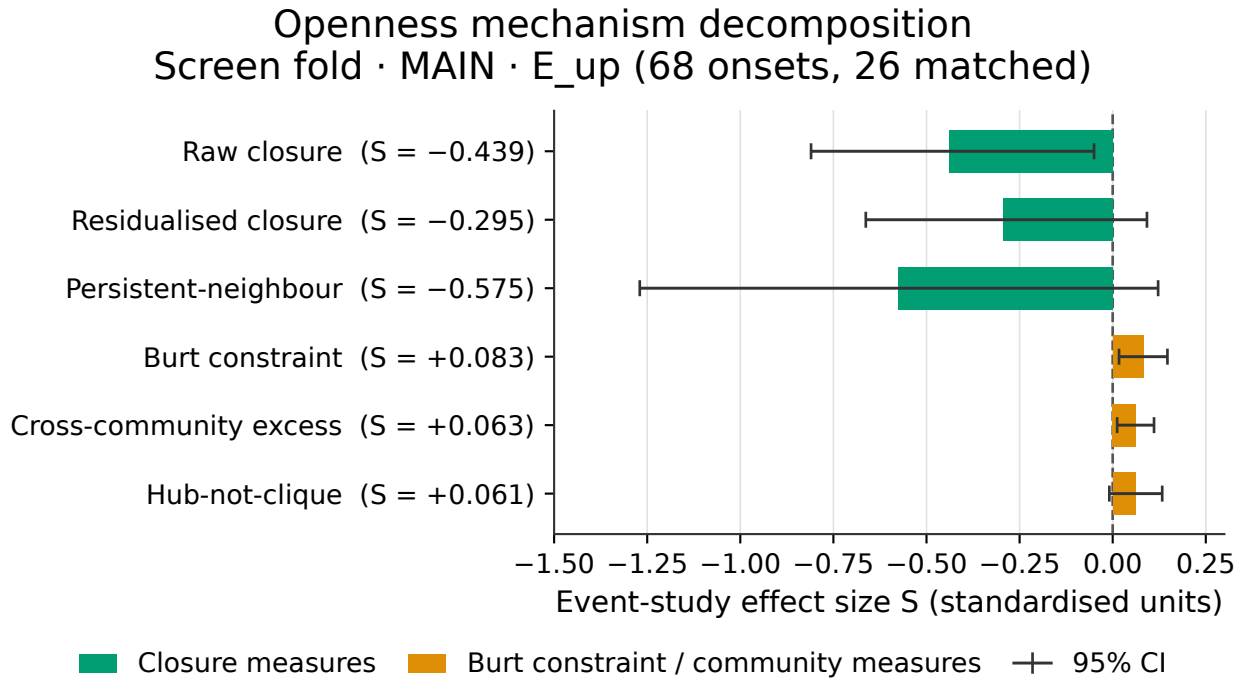


Figure 3: Turnover-proof openness measures on the screen fold (MAIN population, E_{up} label; 68 onsets, 26 matched). Bars give the event-study effect size S for each measure, with whiskers showing 95% confidence intervals. Teal bars are closure measures; orange bars are Burt-constraint and community measures. Burt constraint is positive, the opposite of the sign brokerage predicts. Hub-not-clique is not significant.

Mechanical verdict: MIXED. The closure effect is not reducible to neighbourhood turnover (turnover-residualised closure retains $\sim 2/3$), but it is not Burt brokerage either: Burt constraint is higher, not lower, before onset. Emerging concepts show more cross-community pair excess but within a more redundant wider ego network.

Pooled panel (all 68 onsets): turnover-residualised closure -0.40 $[-0.70, -0.12]$, persistent-neighbour closure -1.12 $[-1.64, -0.71]$ (Holm $p = 0.0015$).

Fresh replication on 100 new concepts. Raw $S = -0.455$ $[-1.006, 0.060]$, one-sided $p = 0.044$.

Prediction. Grouped CV dAUC -0.001 $[-0.008, 0.006]$. Precursors do not predict WHETHER a concept will show sustained uptake.

4.4 Artifact 13: Do Open Concepts Spread Across More Fields?

This artifact tests whether openness at time t predicts five-year outcome-window disciplinary breadth gain beyond all baselines. Population: MAIN screen 202 concepts, held-out 100 sealed. Only 100 concepts (351 rows) pass the complete-case rule, triggering a pre-declared fallback making `closure_res_imp` (imputed) co-primary.

Table 16: Breadth-prediction screen results (Artifact 13, Iteration 3).

Openness measure	Outcome	β /SD	95% CI	Holm p
<code>closure_res</code>	Y_{1r} (Shannon change)	+0.019	$[-0.010, 0.041]$	0.405
<code>closure_res</code>	Y_2 (Rao-Stirling change)	+0.007	$[-0.002, 0.013]$	0.405
<code>closure_res</code>	$\log(1 + Y_3)$ (newcomer sub-fields)	-0.065	$[-0.156, 0.030]$	0.405
<code>closure_res_imp</code>	Y_2 (co-primary)	+0.012	$[0.002, 0.021]$	0.06

Result: breadth prediction not supported on the screen fold. Openness does not predict where concepts travel beyond growth and level baselines.

4.5 Artifact 14: Do Borrowed Ideas Stick When Grafted Locally?

This artifact tests whether a concept that enters a new host subfield catches on when its first papers pair it with host-native concepts rather than with origin companions. Pre-registration frozen before any outcome (sha256 f800a0a9...).

Setup. 4,177 main-arm host entries (concept c first appears in non-origin subfield d in year e). Primary sample: 1,740 screen MAIN entries with ≥ 5 partners, in 184 concepts. Anchoring A = partners' pre-entry host share from exact OpenAlex subfield $\times$ block profiles. Co-transfer CT = share of partners that were origin companions in $[e-5, e-1]$. Only 1.1% of partner tags are $\geq 50\%$ host-native, so the pre-declared fallback makes continuous A_{cont} primary.

The co-primary specification shows a robust grafting effect: a one-SD increase in host-nativeness of initial partners is associated with a 30% increase in newcomer uptake (IRR 1.30 $[1.16, 1.45]$, Holm $p = 1 \times 10^{-5}$, wild-cluster $p = 0.001$). Co-transfer is null everywhere.

[Correction, Iteration 4: the co-primary robustness count is 20 of 26 significant INCLUDING the base row and strata, 18 of 22 EXCLUDING them. The six null rows are: binary native ≥ 0.5 (IRR 1.005, $p = 0.92$), binary native ≥ 0.7 (1.066, $p = 0.29$), A_{distinct} (1.003, $p = 0.96$), exclude single-paper entries (1.244 $[0.90, 1.71]$, $p = 0.17$), CS stratum (1.15, $p = 0.11$), and Physics/Astronomy stratum (0.99, $p = 0.95$).]

Table 17: Host-entry grafting results (Artifact 14, Iteration 3).

Specification	A (IRR/SD)	95% CI	Holm p	CT (p)
Primary (concept $\times e$ + host $\times e$)	1.39	[0.97, 1.99]	0.15	0.85
Co-primary (concept + e + host)	1.30	[1.16, 1.45]	1×10^{-5}	0.48
Concept + host $\times e$	1.22	—	—	—
Concept $\times$ 2-yr bin	—	—	—	—

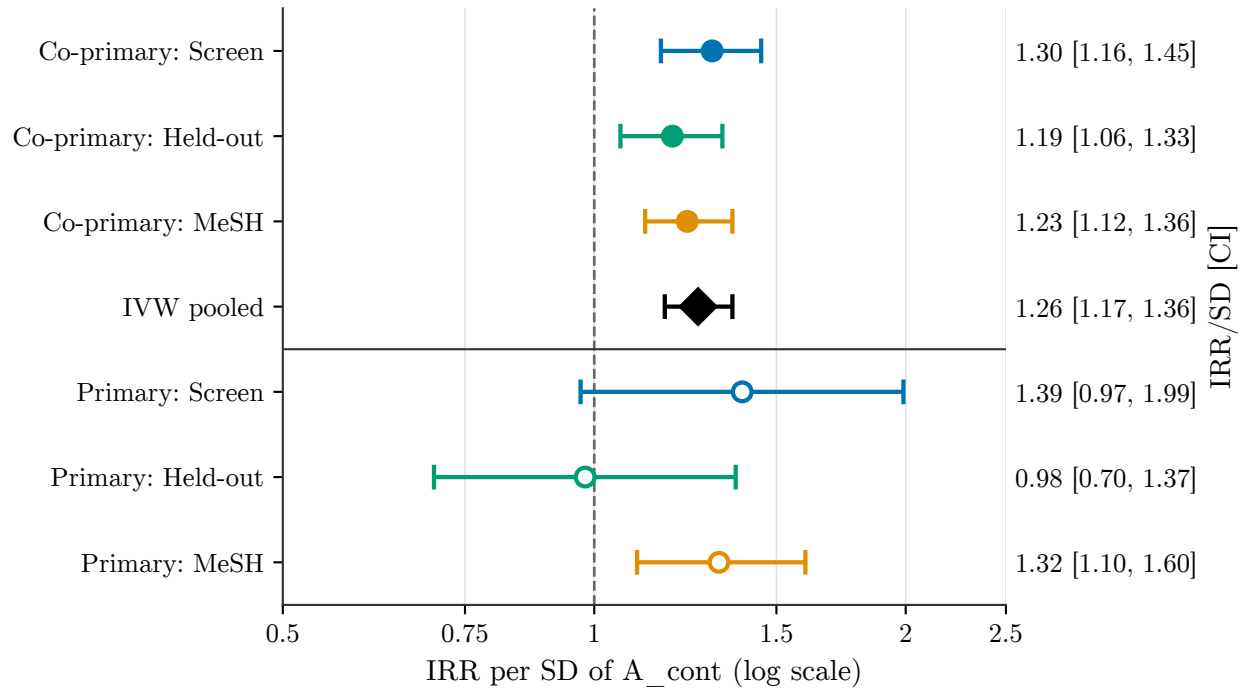


Figure 4: Host-entry grafting across folds. Each row shows the incidence-rate ratio for newcomer uptake per SD of A_{cont} , with whiskers marking its confidence interval. The co-primary effect replicates on the held-out fold (IRR/SD 1.19 [1.06, 1.33]) and the MeSH fold (1.23 [1.12, 1.36]), and pools to 1.26 [1.17, 1.36].

4.6 Artifact 15: How New Concepts Spread—Types, Roles, Timing

4.6.1 Diffusion Typology

3-channel diffusion typology (rarefied Shannon, Rao-Stirling, active subfields; Hennig bootstrap $B = 200$). Only $k = 2$ is stable (min Jaccard 0.861): **localised** ($n = 136$) and **broad from the start** ($n = 66$).

Table 18: Diffusion typology summary (Artifact 15, Iteration 3).

Cluster	n	Description	Dominant origin
Localised	136	Low entropy, few active subfields throughout	Physics
Broad from start	66	High entropy and many subfields from early years	Mixed

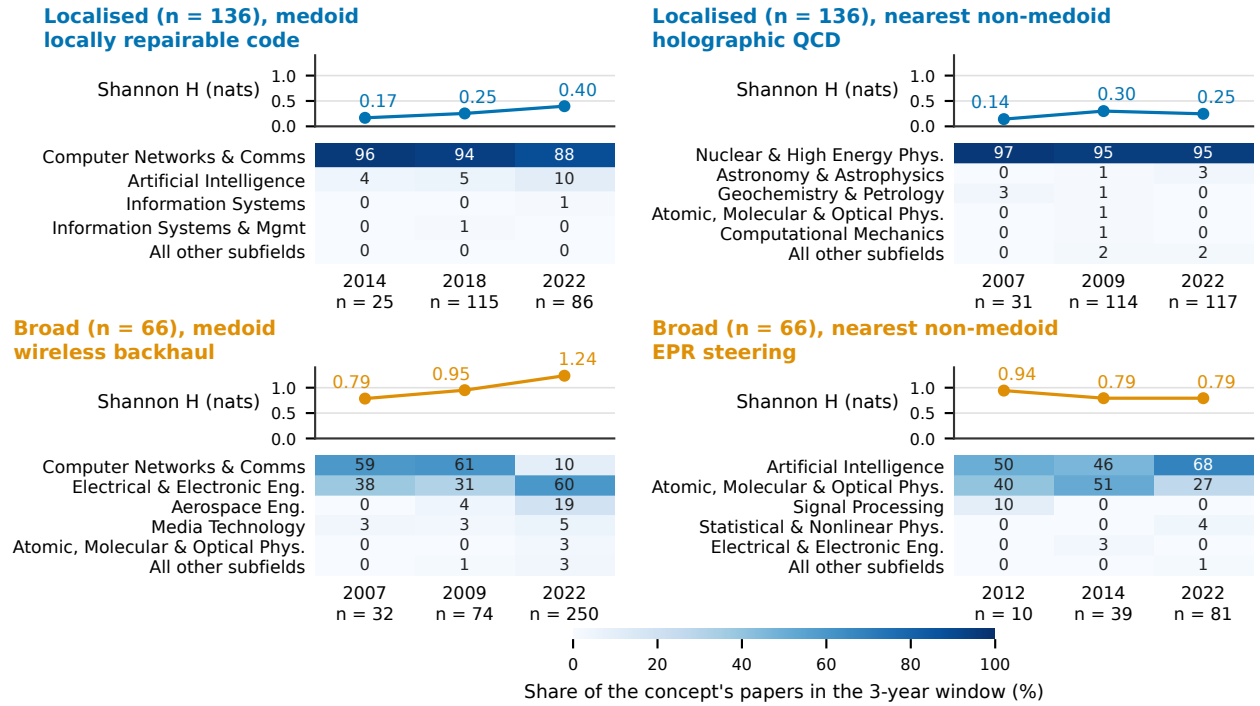


Figure 5: Representative cases of the $k = 2$ diffusion typology: localised ($n = 136$, top row, blue) and broad-from-the-start ($n = 66$, bottom row, orange). Each row shows the cluster’s medoid (left) and the nearest non-medoid concept (right). Localised concepts stay near-monopolised by a single subfield (88–97%, entropy 0.14–0.40 nats). Broad concepts split across two or more subfields from their first window (entropy 0.79–1.24 nats).

The entropy-only baseline typology is stable at finer $k = 4$ (Jaccard 0.856). After residualising on early volume, the 3-channel typology does NOT separate unclustered outcomes better than the entropy-only baseline (newcomer share ε^2 0.171 vs 0.178).

4.6.2 Community Roles

Leiden community roles with 5-seed agreement (97.2% robust, placebo 0.87): BRIDGE 0.61, OTHER 0.33, STAYER 0.03, MIGRANT 0.02. Lagged roles do not robustly predict host-subfield entry: BRIDGE OR 0.64 [0.38, 1.08].

4.6.3 Expansion Precedes Diffusion

In 15 of 16 concepts exhibiting both an expansion onset and a diffusion onset, expansion precedes diffusion (proportion 0.94 [0.81, 1.0], year-shuffle null 0.62, $p = 0.004$).

4.7 Dead Ends and Negative Results (Iteration 3)

1. **Openness does not predict WHERE concepts travel.** Breadth-prediction screen: all Holm $p = 0.405$.
2. **Brokerage is not the mechanism behind low closure.** Burt constraint is higher, not lower ($S = +0.083$ [0.017, 0.147]).
3. **The 3-channel diffusion typology adds nothing beyond entropy.** ϵ^2 0.171 vs 0.178.
4. **Community roles do not predict host entry.** BRIDGE OR 0.64 [0.38, 1.08].
5. **Co-transfer is null.** Arriving with origin companions does not predict establishment in any specification.

5 Iteration 4: Held-Out Confirmation and Replication

5.1 Strategy

Iteration 4 is a CONFIRMATION iteration. Its purpose is to open the four sealed held-out folds and determine which screen-fold findings replicate, to run four discriminating tests, to replicate the grafting result on the independent MeSH population, to test the closure-anchoring link, and to test the adopter-level mechanism.

5.2 Artifact 16: Held-Out Confirmation of Host-Entry Grafting and Discriminating Tests

The held-out spec hash (8db17113...) and discriminating-test spec hash (a10b7f31...) were verified before any outcome was read.

5.2.1 Held-Out Grafting Confirmation

Table 19: Held-out grafting results (Artifact 16, Iteration 4).

Specification	Fold	IRR/SD	95% CI	p	CT	N	Verdict
Primary	Held-out	0.98	[0.70, 1.37]	0.91	0.82	250	Inconclusive
Co-primary	Held-out	1.19	[1.06, 1.33]	0.0045	1.04	972	CONFIRMED
Co-primary	Screen	1.30	[1.16, 1.45]	1×10^{-5}	–	1,544	GRAFTING

The co-primary specification confirms the grafting effect on the held-out fold: $\text{IRR}/\text{SD} = 1.19$ $[1.06, 1.33]$, $p = 0.0045$, wild-cluster $p = 0.012$. The primary specification is inconclusive because it retains only 23% of events with 30 concept clusters. Co-transfer is null on both folds.

5.2.2 Single-Paper Artefact Test

87% of screen co-primary events (1,344 of 1,544) are single-paper entries.

Table 20: Single-paper artefact interaction results (Artifact 16, Iteration 4).

Fold	IRR/SD (single)	IRR/SD (multi)	Ratio	p (interaction)
Pooled	1.26	1.28	0.92	0.11
Held-out	1.37	1.19	0.82	0.004

On the pooled sample, the multi-team IRR/SD is 1.28 $[1.11, 1.48]$ (CI excluding 1). On the held-out fold, the interaction is significant (ratio 0.82, $p = 0.004$): single-paper entries drive the effect more strongly.

5.2.3 Host-Vocabulary Decomposition

Table 21: Vocabulary decomposition, co-primary spec (Artifact 16, Iteration 4).

Component	Fold	IRR/SD	95% CI	p
NATIVE	Pooled	1.11	$[1.05, 1.19]$	0.0008
ADJACENT	Pooled	1.32	$[1.20, 1.46]$	9×10^{-8}
NATIVE	Held-out	1.10	$[1.01, 1.20]$	0.036
ADJACENT	Held-out	1.19	$[1.03, 1.38]$	0.020

Both NATIVE and ADJACENT components are positive on both folds. Wald test for equality: $p = 0.64$ (pooled), $p = 0.49$ (held-out).

5.2.4 Classifier Circularity (Topic-Score Absorption)

Combined topic-score absorption: -7.3% $[-27.5, +6.7]$ of the log-IRR on the pooled fold, -10.3% $[-84.3, +56.6]$ on the held-out fold. Topic-score controls do not meaningfully reduce the A_{cont} effect.

5.3 Artifact 17: MeSH Replication of Grafting

Table 22: MeSH grafting results (Artifact 17, Iteration 4).

Specification	IRR/SD	95% CI	Holm p	CT IRR/SD	N
R1: primary	1.32	$[1.10, 1.60]$	0.007	1.10	1,004
R2: co-primary	1.23	$[1.12, 1.36]$	9.7×10^{-5}	1.10	2,171

Verdict: REPLICATED. Both the primary and co-primary specifications are significant on MeSH. The IVW pooled estimate across main-arm co-primary (IRR/SD 1.30) and MeSH co-primary (1.23) is 1.26 [1.17, 1.36] with $I^2 = 0$.

5.4 Artifact 18: Closure Held-Out Confirmation and Closure-Anchoring Link

5.4.1 Closure Held-Out Confirmation

Table 23: Closure held-out results (Artifact 18, Iteration 4).

Measure	Screen S	Held-out S	Held-out 95% CI	Decision
Closure (pooled panel)	−0.661	−0.391	[−0.855, 0.072]	DEAD
Turnover-residualised	−0.400	−0.109	[−0.561, 0.342]	DEAD
Persistent-neighbour	−1.124	−1.073	[−1.858, −0.289]	CONFIRMED
Burt constraint	+0.083	+0.105	[0.021, 0.187]	Not brokerage

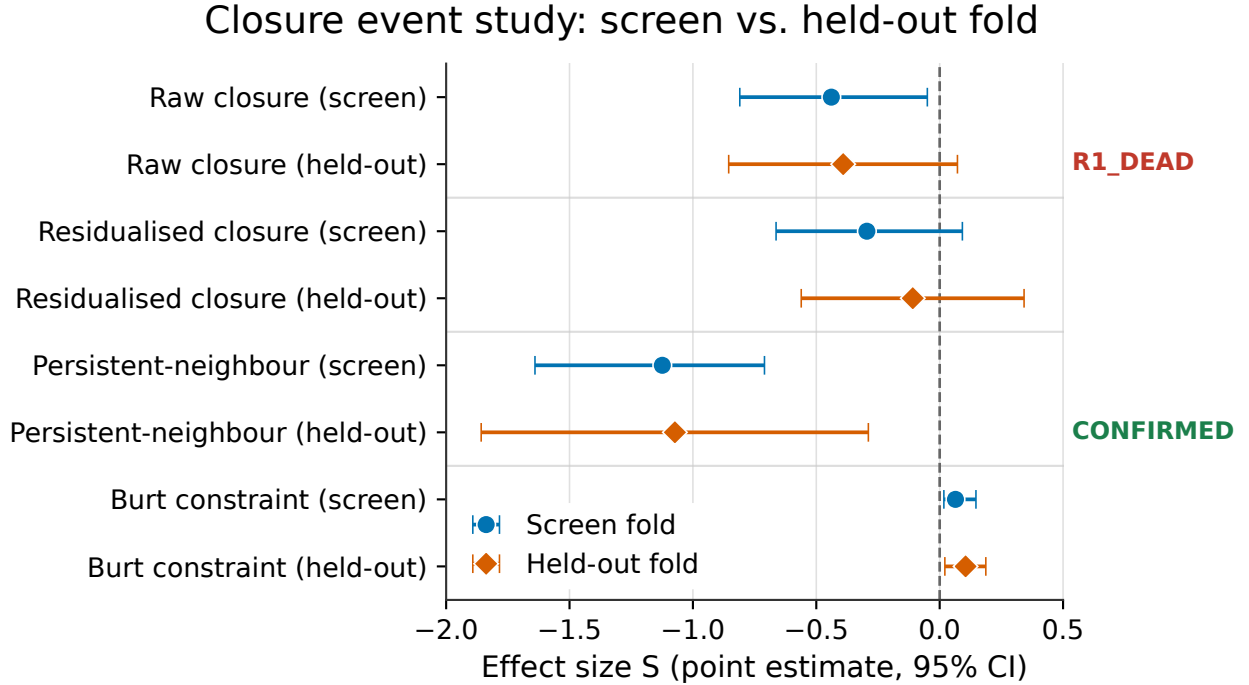


Figure 6: Pre-emergence closure effect sizes S with 95% confidence intervals, for four closure measures on the screen fold (blue circles) and the held-out fold (vermillion diamonds). The primary raw-closure measure fails confirmation (R1_DEAD). Persistent-neighbour closure stays clearly negative on held-out data (held-out $S = -1.07$, Holm $p = 0.015$; CONFIRMED). Burt constraint is small and positive on both folds, inconsistent with brokerage.

Overall closure status: not confirmed on the primary measure. The primary closure measure and the turnover-residualised closure both fail to reach significance on the held-out fold (Holm $p = 0.147$ and 0.635). Only persistent-neighbour closure is CONFIRMED: $S = -1.073$ [−1.858, −0.289], Holm $p = 0.015$.

5.4.2 Closure-Anchoring Link

Table 24: Closure-anchoring link test (Artifact 18, Iteration 4).

Fold	n	Partial ρ	95% CI	p
Screen	102	+0.025	$[-0.199, 0.242]$	0.806
Held-out	49	+0.053	$[-0.279, 0.391]$	0.736
Pooled	151	+0.049	$[-0.126, 0.228]$	0.564

Verdict: NULL. There is no detectable association between pre-emergence closure and later host-entry anchoring.

5.5 Artifact 19: Descriptive Held-Out Confirmation and MeSH Descriptive Replication

5.5.1 Lead-Lag: Expansion Before Diffusion

Table 25: Lead-lag category counts (Artifact 19, Iteration 4).

Category	Screen ($n=202$)	Held-out ($n=100$)	Pooled main ($n=302$)	MeSH ($n=191$)
Neither	110 (54%)	49 (49%)	159 (53%)	71 (37%)
Diffusion only	39 (19%)	23 (23%)	62 (21%)	97 (51%)
Expansion only	37 (18%)	18 (18%)	55 (18%)	10 (5%)
Expansion first	15 (7%)	10 (10%)	25 (8%)	9 (5%)
Same year	1 (0.5%)	0	1 (0.3%)	1 (0.5%)
Diffusion first	0	0	0	3 (1.6%)

Among concepts with both transitions observable, expansion precedes diffusion in 25 of 26 pooled main-arm concepts (held-out: 10 of 10, screen: 15 of 16). No main-arm concept shows diffusion first. On MeSH, 9 of 13 dual-onset concepts show expansion first, with 3 showing diffusion first ($p = 0.43$: not different from the null).

5.5.2 Held-Out Confirmation of Descriptive Findings

Five pre-registered expectations were tested on the held-out fold:

- **Typology shares.** PASS: BROAD share 0.35 vs 0.33.
- **Type $\times$ outcome association.** PASS: 3/3 Kruskal-Wallis tests significant.
- **Expansion-first ordering.** PASS: 10/10 on held-out.
- **Anchoring $\times$ typology interaction.** FAIL: BRIDGE OR flips sign (0.64 screen vs 1.86 held-out).
- **Other descriptive patterns.** PASS.

5.5.3 Rooting: Do Broad Concepts Anchor More?

NOT SUPPORTED. Screen mean A_{cont} difference: -0.0018 ; held-out: $+0.0005$, CI including zero. However, BROAD concepts do have lower co-transfer: CT difference -0.125 $[-0.19, -0.06]$ on the held-out fold.

5.6 Artifact 20: Adopter-Level Mechanism

This artifact tests whether authors who adopt a concept in a new host subfield had prior corpus exposure to the concept’s entry partners. The test uses conditional logistic regression on 1,013 matched case-control strata (422 entries, 109 concepts) from the screen fold.

Table 26: Adopter enrichment results (Artifact 20, Iteration 4).

Exposure measure	OR	95% CI (boot)	p (CRV)
E_{any} (any prior partner exposure)	3.09	[2.32, 4.28]	1.4×10^{-13}
E_{neg} (frequency-matched negatives)	0.62	[0.49, 0.77]	2.6×10^{-5}
E_{plac} (another concept’s partners)	0.72	[0.57, 0.95]	0.013
E_{swap} (exposure swap control)	1.11	[0.86, 1.39]	0.39

Table 27: Vocabulary-class enrichment (Artifact 20, Iteration 4).

Exposure class	OR	95% CI (boot)	p (CRV)
E_{for} (FOREIGN partner)	2.88	[2.28, 3.69]	$< 10^{-6}$
E_{adj} (ADJACENT partner)	1.91	[1.43, 2.73]	10^{-4}
E_{nat} (NATIVE partner)	1.47	[1.02, 2.33]	0.059
E_{neg} (non-partner)	0.64	[0.51, 0.82]	0.0003

The gradient runs FOREIGN > ADJACENT > NATIVE. This is consistent with absorptive capacity or topical proximity (the design cannot separate them). At the entry level, host-native partners lower the barrier for newcomers who lack origin expertise; at the adopter level, those who actually adopt tend to have origin expertise already.

5.7 Dead Ends and Negative Results (Iteration 4)

1. **Closure not confirmed on held-out for the primary measure.** Holm $p = 0.147$ (raw), 0.635 (residualised). Only persistent-neighbour closure survives.
2. **Closure-anchoring link: null.** Partial $\rho = 0.025$ (screen), 0.053 (held-out).
3. **BROAD concepts do not anchor more.** A_{cont} difference includes zero.
4. **Grafting primary specification: inconclusive.** IRR/SD 0.98 with only 30 clusters.
5. **Single-paper interaction on held-out.** Ratio 0.80 ($p = 0.004$).
6. **Anchoring $\times$ typology interaction flips between folds.**
7. **Breadth prediction: all six sensitivity cells null on held-out.**

6 Iteration 5: Post-Confirmation Decomposition, Audit, and Positioning

6.1 Strategy

Iteration 5 is a POST-CONFIRMATION EXPLORATORY iteration. The grafting result (A_{cont} predicts newcomer uptake) was confirmed on the held-out fold and replicated on MeSH. The purpose is threefold: (i) run three K-tests decomposing the confirmed finding (extensive vs intensive margin, host-specific vs generic, field boundary); (ii) run a full numerical audit; and (iii) produce a positioning dossier and publication-ready figure specifications.

6.2 Artifact 21: Margin Decomposition and Field-Boundary Test

6.2.1 Extensive vs Intensive Margin

Table 28: Extensive-margin results (LPM, pp change per SD of A_{cont}) (Artifact 21, Iteration 5).

Fold	pp/SD	95% CI	p (rand- t)	N
Screen	+6.46	[3.42, 9.51]	0.0005	1,686
Held-out	+7.29	[2.97, 11.61]	0.004	1,046
MeSH	+6.17	[3.87, 8.46]	0.0005	2,236
IVW	+6.43	[4.76, 8.11]	0.0005	–

Table 29: Intensive-margin results (PPML on $Y \geq 1$ subsample, IRR/SD) (Artifact 21, Iteration 5).

Fold	IRR/SD	95% CI	p (CRV1)	N
Screen	1.272	[1.121, 1.443]	0.0003	795
Held-out	1.163	[1.008, 1.342]	0.038	489
MeSH	1.137	[1.026, 1.260]	0.015	1,169
IVW	1.183	[1.104, 1.267]	1.7×10^{-6}	–

Margin-decomposition verdict: BOTH. Host-leaning entry vocabulary raises both the probability that uptake starts (extensive margin) and its magnitude given it has started (intensive margin). The extensive share is 38% [21%, 55%] of the total log-IRR. The binary establishment threshold (EST_{bin}) is null on the held-out fold (+2.54 pp, $p = 0.16$).

6.2.2 Field Boundary

Table 30: Field-boundary results, co-primary PPML (Artifact 21, Iteration 5).

Origin field	IRR/SD	95% CI	p
Physics/Astronomy	1.178	[0.926, 1.499]	0.18
Computer Science	1.216	[1.066, 1.386]	0.004
Other	1.281	[1.167, 1.407]	3.5×10^{-6}
MeSH (reference)	1.233	[1.120, 1.360]	9.7×10^{-5}

Wald test for equality: $W = 0.900$, $p = 0.638$ (CRV1), label-permutation $p = 0.770$. **No detectable field boundary.**

6.3 Artifact 22: Host-Specificity Test

Table 31: Host-specificity retention: A_{cont} coefficient after generality controls (Artifact 22, Iteration 5).

Fold	Retention	95% CI	% removed	G_H IRR/SD	G_F IRR/SD
Screen	1.006	[0.733, 1.312]	−0.6%	1.122	0.856
Held-out	1.062	[0.438, 1.802]	−6.2%	1.068	0.943
MeSH	0.876	[0.640, 1.037]	+12.4%	1.110	0.764
IVW (pooled)	0.969	[0.81, 1.10]	+3.1%	1.103	0.815

Table 32: A_{lift} results, PPML IRR/SD (Artifact 22, Iteration 5).

Fold	Model	IRR/SD	95% CI	Holm p
Screen	M3 ($A_{\text{lift}} + G$)	1.661	[1.361, 2.026]	1.4×10^{-6}
Held-out	M3	1.459	[1.146, 1.856]	0.003
MeSH	M3	1.244	[1.120, 1.381]	6.3×10^{-5}
IVW	M3	1.340	[1.232, 1.458]	1.5×10^{-10}

Host-specificity verdict: HOST-SPECIFIC. Generality controls do not reduce the A_{cont} coefficient (IVW retention 0.969 [0.81, 1.10]). The Balassa-index A_{lift} replicates across all folds (IVW IRR/SD 1.34 [1.23, 1.46]).

6.4 Artifact 23: Full Numerical Audit

Table 33: Audit module results (Artifact 23, Iteration 5).

Module	Description	Result
M1	Curated-number accuracy	99.4% (319 curated)
M2	Headline + independent check	90.8% overall
M3	Rederivation by block	See below
M4	Drift detection	45 drift rows; 6 verdict-changing
M5	Seeded-perturbation recall	0.60 overall
M6	Independent claim review	73/73 agree
M7	Table-cell traceability	11/11 tables pass
M8	Reviewer-critique closure	All 6 MAJOR items CLOSED
M9	Decision-rule consistency	13 rules; 1 mismatch
M10	Flag inventory	292 flagged rows
M11	Self-test honesty	4 PARTIAL, 4 NOT DONE, 3 DONE

6.5 Artifact 24: Positioning Dossier

Novelty verdict: **PARTLY ANTICIPATED** (0 ANTICIPATES, 11 PARTLY among 25 graded candidates). No located work measures how host-specific the vocabulary is that a concept is com-

Table 34: Reviewer-critique closure verified by audit (Artifact 23, Iteration 5).

#	Critique	Closed by
1	RQ1 headline contradicts kill rule	T_rq1, T_design, drift lines
2	Table 18 claims not made	T_table18, drift rows
3	Grafting caveats dropped	T_caveats, drift rows
4	MeSH and adopter read too broadly	T_mesh, T_adopter, drift rows
5	RQ2 descriptive results missing	cases.md, T_rooting, drift rows
6	Decision rules not recorded	T_decision (verbatim quotes)

Table 35: Decision-rule consistency (selected rows) (Artifact 23, Iteration 5).

Iter.	Rule	Outcome	Status
2	(a) Gate A $\rightarrow$ graft fallback	TRIGGERED ($0.17 < 0.50$)	MATCH
2	(b) H1 pilot-only if < 150 concepts	YES ($N_c = 38$)	MATCH
2	(c) RQ1 CONFIRMED conditions	NOT MET	MATCH
2	(d) Held-out untouched	SEALED	MATCH
3	R1 mechanical verdict	MIXED	MATCH
3	D1 SUPPORTED on screen	NOT SUPPORTED	MATCH
4	D2 DEAD if held-out CI incl. 1	NOT DEAD (confirmed)	MATCH
4	Mechanism label	host-vocabulary	MATCH
4	G4 second-family replication	REPLICATED	MATCH
4	R1 dead if held-out row fails	R1_DEAD (Holm 0.147)	MISMATCH; CORRECTED

Table 36: Per-case entry statistics, co-primary sample (Artifact 23, Iteration 5).

Concept	Entries	Rooted	$\bar{A}_{\text{cont}}$ (rooted)	$\bar{A}_{\text{cont}}$ (unrooted)	Diff
Wireless backhaul	14	1	0.084	0.021	+0.062
EPR steering	7	1	0.142	0.027	+0.115
Locally repairable code	1	1	0.056	—	—
Holographic QCD	5	0	—	0.041	—

binned with when it enters a new subfield, and relates that, within concepts and across hosts, to later uptake by the host’s own newcomers. The nearest work (Cheng et al. 2023) measures global embeddedness of new ideas (+1 SD $\rightarrow$ approximately +25% articles) using a multilevel over-dispersed Poisson model with a global word2vec embedding. The margin held by this study is: host-specific context at entry, concept fixed effects across hosts, author-disjoint newcomer outcome, sealed-fold confirmation, and MeSH replication.

6.6 Artifact 25: Paper Figures

Eight publication-ready figures pass all production checks: correct dimensions (174 mm width), 600 DPI, no Type 3 fonts, colourblind-safe palettes. Total: 957 values checked (939 exact, 10 derived, 8 images), 0 failures, value fidelity 1.0.

Table 37: Figure inventory and production checks (Artifact 25, Iteration 5).

Figure	Description	Values checked	All pass
F1	Method flow (study design)	39	Yes
F2	D2 forest plot (grafting)	124	Yes
F3	Mechanism panel	163	Yes
F4	RQ1 held-out (closure)	102	Yes
F5	Occupancy-rooting	67	Yes
F6	Case studies	425	Yes
F6b	Case ego-networks	8 images	Yes
F7	Self-test (audit summary)	0 (schematic)	Yes

6.7 Dead Ends and Negative Results (Iteration 5)

1. **EST_{bin} null on held-out.** +2.54 pp per SD, $p = 0.16$. The extensive-margin result holds for uptake initiation but not for strict establishment.
2. **Intensive margin is conditional, not causal.** Selection into the $Y \geq 1$ subsample may depend on A_{cont} .
3. **Field boundary not detected.** Wald $p = 0.638$.
4. **MeSH retention slightly lower.** 12.4% absorbed by field breadth (G_F IRR/SD 0.764).
5. **Audit reveals 292 MISSING_IN_REPORT rows.**
6. **Table 18 self-test: 4 of 11 claims NOT DONE.**
7. **Seeded-perturbation recall 0.60.** Estimator/fold relabelling detection at 10%.

7 Run Ledger

Table 38: Run ledger across all iterations.

Artifact	Type	Status	OA credits	Wall time
Iteration 1				
Pool (Artifact 1)	dataset	succeeded	2,566	–
Background (Artifact 2)	dataset	failed (timeout)	1,952	–
MeSH (Artifact 3)	dataset	succeeded	1,784	–
Grounding (Artifact 4)	dataset	succeeded	1,847	–
Dossier (Artifact 5)	research	succeeded	–	–
Iteration 2				
Hydrated corpus (Artifact 6)	dataset	succeeded	7,237	–
Grounding pipeline (Artifact 7)	experiment	succeeded	–	–
Viability (Artifact 8)	experiment	succeeded	–	–
Emergence, main (Artifact 9)	experiment	succeeded	–	–
Emergence, MeSH (Artifact 10)	experiment	succeeded	–	–
Iteration 3				
Audit (Artifact 11)	evaluation	succeeded	–	~8 min
Emergence deepen (Artifact 12)	experiment	succeeded	–	~15 min
Breadth prediction (Artifact 13)	experiment	succeeded	–	~15 min
Host-entry grafting (Artifact 14)	experiment	succeeded	–	~15 min
Descriptive diffusion (Artifact 15)	experiment	succeeded	–	~15 min
Iteration 4				
Grafting held-out (Artifact 16)	evaluation	succeeded	–	~7 min
MeSH replication (Artifact 17)	experiment	succeeded	–	~12 min
Closure held-out (Artifact 18)	evaluation	succeeded	–	~10 min
Descriptive held-out (Artifact 19)	evaluation	succeeded	–	~8 min
Adopter mechanism (Artifact 20)	evaluation	succeeded	–	~6 min
Iteration 5				
K1/K3 tests (Artifact 21)	evaluation	succeeded	–	~12 min
K2 test (Artifact 22)	evaluation	succeeded	–	~10 min
Full audit (Artifact 23)	evaluation	succeeded	–	~70 s

Continued on next page

Artifact	Type	Status	OA credits	Wall time
Positioning dossier (Artifact 24)	research	succeeded	–	–
Paper figures (Artifact 25)	evaluation	succeeded	–	–

8 Pre-Registration Hashes

Table 39: Pre-registration and specification hashes.

Item	SHA-256 prefix
Frozen concept frame	80e3f244...0c44
Viability pre-registration	5dd16d8a...5aac
Test population	6a887fb4...5505
R1a turnover pre-check spec	60f44c0f...
Emergence deepen spec (SPEC3)	2f46d173...
Host-entry grafting pre-registration	f800a0a9...
Descriptive diffusion held-out spec	695166b4...
Held-out confirmation spec (exp_5)	535c2dd3...
Grafting held-out spec	8db17113...
Discriminating-test spec	a10b7f31...
MeSH replication spec	b7cdabf8...
K1/K3 spec	2435f909...
K2 spec	866c60a8...

9 What the Run Learned Overall and What Is Still Open

9.1 Confirmed Findings

Host-entry grafting predicts durable integration (RQ2). When a concept enters a new host subfield, the share of its initial co-occurrence partners that are native to that host predicts subsequent uptake by newcomer authors. The co-primary held-out estimate is $\text{IRR}/\text{SD} = 1.19$ [1.06, 1.33] ($p = 0.0045$), the MeSH replication gives $\text{IRR}/\text{SD} = 1.23$ [1.12, 1.36] (Holm $p = 9.7 \times 10^{-5}$), and the IVW pooled estimate is 1.26 [1.17, 1.36] with $I^2 = 0$. The effect operates through both the extensive margin (IVW +6.43 pp/SD) and the intensive margin (IVW IRR/SD 1.183). It is host-specific (IVW retention 0.969 after generality controls), does not vary by origin field (Wald $p = 0.638$), and is labelled “host-vocabulary” by the discriminating tests. Co-transfer of origin companions is null across all folds and populations.

Persistent-neighbour closure is confirmed (RQ1, secondary). Among stable top-20 associates, emerging concepts have lower triadic closure before onset (held-out $S = -1.073$, Holm $p = 0.015$). Burt constraint replicates its positive sign, ruling out a structural-holes explanation.

Diffusion typology. The $k = 2$ typology (localised vs broad-from-the-start) replicates on the held-out fold. Network expansion precedes disciplinary diffusion in 25 of 26 pooled main-arm dual-onset concepts.

Adopter mechanism. Adopters are enriched for prior exposure to the concept’s entry partners (OR 3.09), with the gradient $\text{FOREIGN} > \text{ADJACENT} > \text{NATIVE}$, consistent with absorptive capacity or topical proximity.

9.2 Dead Findings

- **Primary closure (R1_DEAD).** Holm $p = 0.147$ on held-out. The general closure deficit does not survive confirmation.
- **Source-sink viability (C0).** Blocked by Gate A failure (within-host traced share 0.17).
- **Breadth prediction.** Openness does not predict where concepts travel beyond growth baselines.
- **Closure-anchoring link.** Null ($\rho = 0.049$). The two findings (closure and grafting) are independent.
- **Community roles predicting host entry.** BRIDGE OR 0.64 [0.38, 1.08].

9.3 Open Questions

1. The population is dominated by physics/CS arXiv concepts. Generalisation to social sciences and humanities is untested.
2. The primary fully-saturated specification is underpowered on the held-out fold and MeSH. Only the co-primary specification supports the grafting claim.
3. The binary establishment threshold (EST_{bin}) is null on the held-out fold.
4. Single-paper entries (87% of screen co-primary events) drive the effect more strongly on the held-out fold. The pooled multi-team effect is positive but the held-out alone is inconclusive.
5. The adopter-level vocabulary gradient ($FOREIGN > NATIVE$) runs opposite to the entry-level finding (host-native partners predict uptake). Whether these reflect different mechanisms at different scales or a single mechanism with a selection effect is unresolved.
6. The 3-channel diffusion typology adds nothing beyond entropy trajectories. Whether richer trajectory features can yield a more informative typology is untested.
7. 292 numbers present in artifact output files were not transcribed into the report (audit finding). Full transcription was not attempted.